

Shorts Studio

The complete guide

You do not need to know how to code to use this. If you can copy and paste, you can run it.

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This guide covers everything: what the tool is, how to start it, how to make a video, how to keep your work safe with Git, and how to run the same tool on a second computer.

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1. What this tool actually is

You pick a subject, press a button, and about three minutes later you have a finished video with a real human-sounding voiceover, ready to upload. Or you write the script yourself and let the tool do the rest — see Writing the script yourself.

Four services do the work, and they all run from one page in your browser:

	What it does	Needed?
Google Gemini	Writes the question, the four options, the explanation and the exact words the narrator says — or, in explainer mode, the whole storyboard. Also writes your title, tags and description at the end.	Required
ElevenLabs	Turns that script into speech.	Required
DeepSeek	Solves the question independently and says whether it agrees with Gemini.	Optional
Pexels + NASA	Free photos to sit behind the text. NASA needs no key.	Optional

The video itself is drawn on your own computer by **Remotion**, which is why rendering costs nothing.

Two ways to tell it

- **Quiz** — a question, four options, a countdown and the reveal. The format that stops a scroll.

- **Explainer** — no question at all. A storyboard that builds understanding scene by scene using analogies and diagrams: a title card, an analogy, a labelled diagram, the steps, a comparison, a timeline, and a recap. Aimed at 3 to 5 minutes in 16:9.

Two kinds of video

- **Curiosity STEM** — counter-intuitive science and maths for a general audience.
- **Electrical exam prep** — in the style of a real paper, aimed at a specific exam: GATE EE, ESE/IES, SSC JE, RRB JE, State AE/JE, PSU (UPPCL/DMRC/NTPC/BHEL), or ITI/Wireman.

Two shapes

- **Portrait 9:16** — Shorts, Reels, TikTok. 30 to 90 seconds.
- **Landscape 16:9** — a proper explainer for YouTube. 2 to 5 minutes.

How long does one video take? Roughly 3–4 minutes of your attention, most of it waiting. The very first video takes longer, because the tool downloads its rendering engine once.

2. Setting up, once

Node.js

This is the program that runs everything. Check whether you already have it — press the **Windows key**, type `powershell`, press **Enter**, then type:

```
node -v
```

A reply like `v20.11.0` or higher means you are fine. An error means you need it: go to nodejs.org, download the big green **LTS** button, click Next through the installer, then **close and reopen PowerShell**.

Your API keys

An "API key" is a long password that lets this tool use an online service on your behalf.

Gemini (required — writes the questions)

1. Go to <https://aistudio.google.com/app/apikey>
2. Sign in with any Google account → **Create API key** → copy it.
3. It looks like `AIzaSyD...`

ElevenLabs (required — does the voice)

1. Go to <https://elevenlabs.io> and make a free account.
2. Go to <https://elevenlabs.io/app/settings/api-keys> → **Create API key** → copy it.
3. It looks like `sk_1a2b3c...`

DeepSeek (optional — double-checks the answer)

1. Go to https://platform.deepseek.com/api_keys → **Create new API key** → copy it.

2. It looks like `sk-1a2b3c...`

Worth adding. Gemini writes the question *and* marks its own answer, so nothing catches a confidently wrong one. A check costs a fraction of a cent.

Pexels (optional — backdrop photos)

1. Go to <https://www.pexels.com/api/> → sign up free → **Get Started** → copy the key.

Without it you still get NASA's public-domain library, which is excellent for space and physics but thin for chemistry and biology.

Keep these private. Anybody holding your key can spend your credits. Never put one in a screenshot, a video, or a message. They are stored only in your browser, never in a file — which is also why they are never uploaded to GitHub.

Install the packages

Open PowerShell and go to the project:

```
cd C:\Projects\shorts-studio
```

Then:

```
npm install
```

This takes one to three minutes. Yellow warnings are normal — only a red **ERR!** means trouble.

If you see "running scripts is disabled on this system"

This is a Windows security default, not a fault in the tool. Two ways past it.

The easy way, changing nothing — add `.cmd`:

```
npm.cmd install
```

Use `npm.cmd` everywhere this guide says `npm`. That is the whole fix.

The permanent way — run this once and press Y:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

`RemoteSigned` lets scripts written on your own computer run, while anything downloaded from the internet must still be signed. It applies to your account only, and it is what Microsoft recommends for people who develop on Windows.

3. Starting it, every time

```
cd C:\Projects\shorts-studio
```

```
npm start
```

Two things start together: the **helper**, which talks to Gemini and ElevenLabs and does the rendering, and the **web page** you actually use. The helper opens your browser once it is ready, so give it a few seconds. If it does not appear, paste this in yourself:

```
http://localhost:5173/
```

That address never changes. If something else is already using it the tool stops and says so, rather than quietly moving to another one.

"Starting up..." for a second or two is normal. The page loads faster than the helper does, so on a cold start it waits for it and tells you. It clears by itself. Only if it is still trying after twenty seconds will it tell you the helper is genuinely missing.

Leave the PowerShell window open the whole time. That black window *is* the engine. Closing it stops the tool. When you are finished for the day, click it and press `Ctrl + C`.

4. Making a video — the seven steps

Seven numbered buttons run across the top. You move left to right. Steps you have not earned yet stay greyed out on purpose, so you cannot get lost.

Step 1 — Keys

Paste your keys and press **Test my keys**. You want green ticks. It also reports how many ElevenLabs voice credits you have left this month.

You do this once, ever. If a key is rejected, the cause is almost always a stray space — clear the box and paste again.

Step 2 — Topic

Where you decide what the video is about.

Setting	What it does
How it is told	Quiz or Explainer . Choosing Explainer switches to 16:9 and at least three minutes, because that is what the format needs.
What kind of video	Curiosity STEM for a general audience, or Electrical exam prep for candidates revising.
Exam <i>(exam prep only)</i>	Sets the depth. GATE and ESE want derivation and analysis; SSC and RRB want standard formulas at speed; ITI wants practical wiring and safety with no calculus.
Format	Portrait or landscape. Pick this first — it changes how much script is written and how the frame is laid out.
Subject	The broad area. In exam mode this is the syllabus section.
Sub-topic	A dropdown of suggestions for that subject — 38 for Power Generation, for example. Pick one to concentrate a run of videos on a single section, or leave it on <i>Any</i> .
Or type your own topic	Free text. Whatever is in this box is what gets used; clear it to let Gemini choose.

Setting	What it does
Who is watching	Sets vocabulary and assumed background.
Difficulty	<i>Very easy</i> through <i>Brutal</i> .
Question style	<i>Mathematical</i> = the viewer must calculate. <i>Theoretical</i> = they must reason. <i>Real-world</i> = anchored in daily life.
Narration language	The language spoken <i>and</i> shown. Kannada, Hindi, Tamil and Telugu render in their own script — pair with a matching voice in step 4 and the Multilingual v2 model.
Curiosity factor	The most important dial. 8 or 9 is the sweet spot: counter-intuitive enough to make people comment.
Target length	40–50s performs best for Shorts. 180–240s is the sweet spot for explainers.
How many diagrams	<i>Rich</i> (the default) puts a chart, circuit or animation on nearly every scene. <i>Balanced</i> on most explanation scenes. <i>Sparse</i> only where it really helps.
Gemini model	Flash is fast and cheap. Pro is worth it for hard maths. The list is built from your own key, so it only offers models you can actually use.

Curiosity high — what is trending now

Press **What is trending now?** and Gemini *searches the live web* — it is not answering from memory — then judges which of what it finds would actually make a good video. You get eight ideas, each with why it is being talked about and the counter-intuitive angle to build around. Click one and it fills the topic box.

It searches your domain: science and technology in curiosity mode, or grid, generation, storage and standards news in electrical mode. **Where this came from** lists the pages it actually read, so you can check a claim yourself.

Trending is not the same as true. A story spreading fast is exactly the kind that turns out to be half right. Verify the answer in step 3, and run the DeepSeek check if you have a key.

This needs a **2.5 model** (Flash or Pro) — older models cannot search. It costs one Gemini request.

Your intro (optional). Type a greeting — "*Hi, it's Hemanth here. Ready for today's question?*" — and it becomes the first thing the video says. There are one-click presets. **Gemini never rewrites this:** whatever you type is spoken and shown word for word.

Press **Generate the question.** Five to twenty seconds.

Step 3 — Script

You check Gemini's homework. **This is the most important step and the one people skip.**

Only the spoken words appear on screen. The narration *is* the on-screen text — shown a few words at a time, each word lighting up as it is said. There is no separate headline to keep in step, and no labels, step counters or subject chips cluttering the frame.

Every line is editable. **Read the question and satisfy yourself the marked answer is right** — AI is confidently wrong sometimes, and ten seconds here saves publishing something embarrassing.

The second opinion. With a DeepSeek key you get a **Check the answer** button:

Badge	Meaning
✓ DeepSeek agrees	Both models got the same answer.
⚠ Reservations	The answer stands, but something needs a look.
✗ DeepSeek disagrees	They got different answers . One is wrong.

On a disagreement you get a one-click button to switch the marked answer — but read the reasoning first. DeepSeek is not automatically right either; the value is knowing to look. Pick **Reasoner** for hard maths.

Edit the question afterwards and the badge is replaced by "*the check is out of date*" — a stale tick is never left pretending the new version was verified.

Watch the length. A bar shows how many seconds you have written against the length you asked for. If a generated script came out well under target you also get a warning — regenerate now, while it is still free.

The bar is an estimate at the voice's own speaking pace. **You cannot set how long a scene lasts**, and there is no slider for it anywhere. A scene lasts exactly as long as its recorded narration — that is the whole reason the words on screen stay locked to the voice. The real length arrives with the voiceover in step 4.

Rearranging the video. Every scene has controls in its header:

Control	What it does
The kind dropdown	What the scene is for — hook, question, options, countdown, answer, explanation, outro.
↑ ↓	Move the scene earlier or later.
×	Delete it.
+ Add a scene	Adds an explanation scene before the outro.

Changing any of these means the voiceover has to be recorded again, and the page tells you so if you have already made one.

On an explainer, a scene that draws a layout shows a  **Drawn on screen** row listing every label in it. A label marked ⚠ is one your narration never says — the reveals are timed by matching the spoken words against those labels, so an unmentioned label can only appear on a guess. Mention it in that scene's narration, in the order shown, and the warning clears.

Step 4 — Voice

Pick a voice from your own ElevenLabs account and press **Hear this voice** to sample it. If the list is empty, add any voice from the [Voice Library and reload](#).

The cost is shown before you spend anything. Press **Make the voiceover**.

This is where sync happens. Every clip is decoded in your browser and measured for real — exact length, where sound starts, where it stops. Each scene is then made exactly as long as its own clip, always rounding *up*, so a line can never be cut off. The subtitles are nudged by the few milliseconds of silence every MP3 carries at its start, which lands the highlight on the right syllable.

The options light up as they are read, because the narration reads them in order.

Open **Show the sync report** to see the numbers per scene. If a line ever sounds clipped, turn off **Trim trailing silence** in step 5.

Step 5 — Look

Everything here is instant, free, and never touches the voiceover.

- **Dark or light**, and four layouts: **Simple** (clean), **Elegant** (serif, documentary), **Nerdy** (terminal green on graph paper), **Flashy** (loud, best in a feed).
- **Highlight colour, thinking time** (3–5s), **breathing room**.
- **Show the spoken words** — the read-along text. Leave it on; most people watch on mute.
- **Draw the diagrams** — on the explanation and outro scenes, and a **setup diagram on the question scene**: the circuit, the apparatus or the geometry being asked about. Anything that could hint at the answer is stripped from that one automatically — no charts, no pies, no comparison panels, and no caption. As well as static ones (a formula box, comparison bars, a side-by-side panel, an icon) Gemini can choose a **live animation** from a fixed library of ten: wave interference, a travelling wave, orbits, a projectile arc, a pendulum, a vector field, spreading particles, a graph being drawn, an atom, light refracting — plus six built for electrical and power work: a **circuit** (series or parallel), a **phasor diagram** and power triangle, an AC **waveform** (phase shift, rectified, PWM), a **block flow** that lights up stage by stage (boiler → turbine → condenser → pump), a **transformer** with turns ratio, and a **pie** for a fuel mix or a loss breakdown.
- **Drift topic symbols** — faint themed emoji behind everything.
- **Moving backdrop** — one of thirty slow animations under the whole video. See Moving backdrops below.
- **Backdrop photos** — press **Find backdrop photos** and it searches Pexels and NASA per scene, using a search term Gemini wrote for that scene. **Nothing is applied for you**: a photo library will cheerfully return a beach for "gravity". Click the ones that fit, skip the rest.
- **Sound** — three built-in music beds (calm, tense, upbeat), or load your own file. The music **ducks automatically** under the narration. Effects: a countdown tick, an option whoosh, an answer chime, and a sweep between scenes.

Play the phone preview before rendering. Fixing something here takes a second; after a render it takes minutes.

Step 6 — Export

Choose a quality (**Normal** is right almost always) and press **Render the video**.

The first render is the slow one. The very first time, the tool downloads a rendering browser of about 150 MB. It can look like nothing is happening for several minutes. It only ever happens once per computer.

A 45-second video takes one to three minutes. You get a player, a download button, and the file is saved into the **out** folder automatically.

Step 7 — Publish

Gemini writes your upload kit from the finished video: **several title options** to choose from, a description, tags, hashtags, suggested thumbnail text, and a pinned comment. Each has a copy button.

For exam-prep videos the title and first line of the description lead with the exam name, subject and topic, because that is what people actually type into search.

Writing the script yourself

You do not have to let Gemini write it. On step 2, set the **format** and the **target length**, then press  **Write it myself** instead of Generate.

That drops you straight into step 3 with an empty script of the right shape — the correct beats in the correct order, and about the right number of them for the length you chose. A 180-second explainer gives you 18 scenes to fill in; a 45-second quiz gives you the question, the four options, the countdown, the answer and a few explanation cards.

It costs nothing and needs no Gemini key. Credits are only ever spent on step 4, the voice.

In step 3 every scene has a **kind** dropdown, ↑ ↓ to reorder and × to delete, and there is an **Add a scene** button underneath. A bar shows how many seconds you have written against the length you asked for.

One thing that is not a slider. You cannot set how long a scene lasts. A scene lasts exactly as long as its recorded narration — that is the whole reason the words on screen stay locked to the voice. The bar in step 3 is an estimate at the voice's own speaking pace; the real length arrives with the voiceover in step 4.

Moving backdrops

Step 5 has a **Moving backdrop** — a slow animation under everything, so a scene reads as produced rather than as text on a flat colour. Thirty of them, in ten families: drifting particles, flow fields, constellations, waves, orbits, a circuit-board grid, light beams, spirals, equaliser bars and contour maps.

Auto picks one from your subject: a circuit board for electronics, a lattice for transmission, equaliser bars for power generation, a galaxy for astronomy, contours for climate.

It is capped well below the text at every setting, so it can never make a caption hard to read. Turning it up makes it busier, not more distracting. It costs roughly **20% more render time** with the heaviest of them; set it to *None* if you want the fastest possible render.

A note on units

Voice models read **10 MW** as "ten mili wag". Any unit symbol with a number in front of it — MW, kV, kWh, MVA, Hz, Ω, °C, % and the rest — is expanded for the **voice only**. The narrator says "ten megawatts"; the screen still shows 10 MW, and the highlight lands on it at the right moment.

A symbol with no number before it is left alone, so "A transformer" stays an article and does not become "ampere transformer". Write units the way you normally would.

5. Git — your undo button

Git takes **snapshots** of your project. It is the reason a mistake can never cost you more than a few minutes.

Three ideas, and that is genuinely all you need:

- **A commit** is a labelled snapshot. "Here is what everything looked like at 4pm."
- **Your history** is the list of those snapshots.
- **You can always go back** to any of them.

Git deliberately ignores three things: `node_modules` (hundreds of megabytes, rebuilt by `npm install`), your finished videos, and generated audio. Everything that matters is tracked.

The only commands you need

See what you have changed:

```
git status
```

Stage everything you changed:

```
git add -A
```

Save the snapshot:

```
git commit -m "describe what you changed"
```

The message is for future-you. "Made the countdown longer" beats "update".

See your history:

```
git log --oneline
```

Undoing things

Throw away changes to one file you have messed up:

```
git restore path/to/file
```

Throw away **all** uncommitted changes and go back to your last snapshot:

```
git restore .
```

⚠ That last one is not itself undoable. It discards everything you have changed since your last commit. Commit often and it is never frightening.

6. GitHub — your backup and your bridge

GitHub is a website that stores a copy of your project in the cloud. It does two jobs: it is a backup if your laptop dies, and it is how the same project reaches a second computer.

Your project lives at github.com/HHHkumar/shorts-studio, and it is **private** — only you can see it.

Send your latest snapshots up:

```
git push
```

Bring down changes made on another machine:

```
git pull
```

The rhythm that keeps it painless

Pull before you start. Push before you stop.

A full session looks like this:

```
git pull
```

...do your work, then...

```
git add -A
```

```
git commit -m "what you changed"
```

```
git push
```

If you forget and edit the same file on both machines, Git will ask you to reconcile the two versions. Irritating, but nothing is ever lost.

If a push fails with "Permission denied to <some other name>", Windows has a different GitHub account saved. That is exactly why your project's address includes your username:
`https://HHHkumar@github.com/HHHkumar/shorts-studio.git`. It tells Git which account to use.

7. Running it on another computer

Everything you need is on GitHub, so this is four commands.

1. Open PowerShell where you want the project to live, then:

```
git clone https://HHHkumar@github.com/HHHkumar/shorts-studio.git
```

2. Go into it:

```
cd shorts-studio
```

3. Install the packages (this is why `node_modules` is not in the repo — it is rebuilt here):

```
npm install
```

4. Start it:

```
npm start
```

Three things that do not travel, by design

- **Your API keys.** They live in your browser, never in a file. Paste them into step 1 again.
- **The rendering browser.** The first render on the new machine downloads it again (~150 MB, once).
- **Your finished videos.** They are too big for a repo. They stay on the machine that made them.

After that, both computers are equal. Pull before you start, push before you stop, and they stay in step.

8. What it costs

Service	Free allowance	One video uses
Gemini	A generous free tier	One or two requests. Realistically free.
ElevenLabs	10,000 characters a month	A short uses 500–900 characters (10–20 a month free). A five-minute explainer uses about 4,400 (two a month).
DeepSeek <i>(optional)</i>	Pay as you go, no free tier	A fraction of a cent per check.
Pexels / NASA <i>(optional)</i>	Free	Nothing.
Rendering	Unlimited	Your own computer. Costs electricity.

Changing the look, re-rendering, editing text and picking photos all cost **nothing**, and neither does **Write it myself** — that path needs no Gemini key at all. Only *Generate the question*, *Make the voiceover*, *Check the answer* and *Publish* spend anything.

To stretch ElevenLabs credits: shorter targets, and the **Flash** voice model.

9. When something goes wrong

What you see	What it means	What to do
"running scripts is disabled on this system"	Windows blocks npm's launcher by default.	Use <code>npm.cmd</code> , or run <code>Set-ExecutionPolicy -Scope CurrentUser RemoteSigned</code> once.
Red box: "The helper server is not running"	The helper really is not there — the page waited twenty seconds first.	Reopen PowerShell, <code>cd</code> to the folder, <code>npm start</code> , reload the page.
"Port 3030 is already taken"	A second copy of the tool is still running.	Close the other PowerShell window. The message prints the command to force it if you cannot find it.
"Port 5173 is already in use"	Same thing, for the web half.	As above — close the other window and start again.
"This model spent its whole budget thinking"	A newer Flash model reasoned until it had no room left to write.	Pick a 2.5 model in the dropdown. They are the reliable choice for long scripts.

What you see	What it means	What to do
"No reply after 240 seconds"	The model stalled.	Try again, or switch to a Flash model.
"That ... API key was rejected"	Bad key, usually a stray space.	Re-copy it from the provider and paste again.
"Your ElevenLabs character quota is used up"	Out of voice credits this month.	Wait for the reset, shorten the video, or upgrade.
"Your DeepSeek account has no credit left"	DeepSeek has no free tier.	Top up, or clear the key to turn the check off.
"Gemini rate limit hit"	You generated too fast.	Wait a minute and try again.
"Your key cannot use that Gemini model"	The model is not available to your key.	Reload the page — the dropdown rebuilds from your own key.
No voices in the dropdown	Your ElevenLabs account has no voices saved.	Add one from the Voice Library, reload.
Render stuck at 0% the first time	It is downloading the rendering browser.	Wait. It happens once per computer.
"The output file is locked"	A video player still has the last render open.	Close it and render again.
A spoken line sounds clipped	Trailing silence trimmed too aggressively.	Step 5 → turn off Trim trailing silence .
A diagram shows wrong numbers	Gemini invented them.	Fix that scene in step 3, or turn off Draw the diagrams .
The script is far shorter than asked	Gemini underwrote it.	Step 3 warns you. Regenerate, or switch to a stronger model.
Push fails: "Permission denied to ..."	Windows has another GitHub account saved.	Make sure the address includes HHHkumar@.
A change seems to have no effect	An old server is still running from before.	Ctrl + C in PowerShell, then npm start again.
Everything is confusing	—	Step 7 → Reset everything , then start from step 1.

10. Where things are saved

Inside C:\Projects\shorts-studio:

- out\ — your finished videos. **This is the folder you want.** Nothing here is ever deleted automatically, and nothing here goes to GitHub.
- public\generated\ — voiceover clips and downloaded photos. Cleared automatically after a day.
- public\audio\ — the music beds and effects, regenerated on first boot.
- .git\ — your snapshots. Do not touch it; that is Git's business.
- Your keys and settings — in your browser, not in any file.

11. Questions people ask

Can I use my own voice? Yes. Clone it on the ElevenLabs website and it appears in the step 4 dropdown.

Can I change the fonts and colours? Yes. `src\lib\theme.ts` holds every visual choice, with comments. Save it and the preview updates instantly. If you break it, `git restore src/lib/theme.ts` puts it back.

Does it work offline? No. Gemini, ElevenLabs and the one-time renderer download all need the internet. Re-rendering a video you already generated works offline.

Is my data going anywhere? Your topic settings go to Google, your script to ElevenLabs, and your question to DeepSeek if you enabled it. Nothing else leaves the computer — the helper server listens only on `127.0.0.1`, so nothing on your network can reach it.

Why is my project not in OneDrive any more? OneDrive was syncing `node_modules` — tens of thousands of files — which is slow and can lock files mid-render. Git does the job properly, so the project moved to `C:\Projects`.

Can I sell the videos I make? Check each service's terms. Note in particular that **Remotion is free for individuals and small teams but needs a paid company licence beyond that** — see <https://remotion.dev/license>.

Something is still wrong. Look at the PowerShell window. The last few red lines usually say plainly what failed.

12. A checklist for good videos

- Curiosity factor at **8 or 9**. Boring questions do not get watched.
- 40–50 seconds for Shorts; 180–240 for explainers.
- **Read the question in step 3 and verify the answer yourself.**
- Run the DeepSeek check — a disagreement between two models is the cheapest bug report you will get.
- Keep the read-along text on; most viewers are on mute.
- Only pick backdrop photos that genuinely fit. An unrelated one makes it look worse, not better.
- Thinking time of 3–5 seconds. Longer and people scroll away.
- **Commit and push when you finish.** It takes ten seconds and it is the whole safety net.